CERTIFYING A WORK OF ART

without galleries, without intermediaries

The SHA-256 + CIE Protocol for Artists and Dealers

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Abstract

This paper presents a complete operational protocol for the cryptographic certification of artworks, designed specifically for independent artists and dealers operating in the primary and secondary market outside the institutional gallery circuit. The protocol applies to photographic works on paper, digital illustrations on paper, and traditional paintings: for the first two categories it provides maximum documentary robustness through the cryptographic link between the digital source file and the physical print; for paintings and sculptures, where no digital master exists, a signed photograph of the work constitutes the anchor point of the chain.

The system integrates five technological layers — PDF/A certificate, SHA-256 or BLAKE2b hash, SHA Document, CIE digital signature, PEC certified timestamp — into a chain of authentication verifiable by anyone with access to standard tools, without intermediaries, with legal validity in the European Union under the eIDAS Regulation, in the terms described in Part 3.

The reference context is the Italian independent primary art market, where the vast majority of transactions occur without any verifiable documentation. The exact scale of this phenomenon is structurally impossible to quantify: undocumented transactions leave no trace. Its extent is nevertheless recognised by industry practitioners, archivists, and artists themselves as one of the most widespread weaknesses of the independent primary market. The protocol addresses this gap not by creating new platforms or infrastructure, but by making existing infrastructure accessible: state digital signatures, ISO standard cryptographic algorithms, PDFs verifiable with free software.

The practical validity of the system is documented by the pilot case included in this paper: the donation of "Il Ticket Infinito" to Pietro Mingione, Station Gallery, on 17 March 2026, constitutes the first real and verifiable application of the protocol, with CIE signature on both documents and direct delivery to a professional art market operator.

Scope of application. The protocol was developed and tested on photographic works and digital illustrations printed on paper. It is applicable with the same procedures to traditional paintings and drawings, with the difference that in the absence of a digital source file the signed photograph of the work replaces the master hash. It has not been tested on large-scale sculptures, installations, or ephemeral works: these contexts may require specific adaptations not described in this paper.

This paper distinguishes itself from blockchain and NFT solutions for three structural reasons: (1) it anchors the signer's identity to the state registry through the Electronic Identity Card; (2) it requires no third-party platforms, tokens, or fees; (3) it is verifiable with universally available free tools such as Adobe Acrobat Reader.

Theoretical Preface: The Documentary Vacuum and the Failure of NFTs

The Problem

The independent primary art market in Italy — direct donations, exchanges between artists, sales outside galleries — lacks documentary standards. The scale of this phenomenon is structurally impossible to quantify precisely: undocumented transactions leave no statistical trace. Its extent is nonetheless recognisable indirectly: the absence of shared standards, the lack of documentary obligations for the independent primary market, and the direct testimony of artists, dealers, and archivists all converge in describing non-documentation as the norm, not the exception.

Traditional galleries address this problem with their institutional reputation. But this system structurally excludes the independent artist and creates permanent dependence on the intermediary.

The Failure of NFTs

→ An NFT is a blockchain token certifying ownership of a digital pointer. It does not certify the creator's identity — an Ethereum wallet does not say who you are.

→ Public blockchains are not anchored to state identity. Anyone can create a wallet and mint an NFT in the name of another person's work.

→ Verification requires technical expertise and specific tools. It is not accessible to a traditional collector.

→ NFT platforms charge fees, require registration, and may close.

A Historical Note

Before the advent of digital signatures, the most widespread informal practice among Italian artists for establishing the creation date of a work was the "self-addressed registered letter": mailing documentation of the work to oneself, having the envelope stamped at the post office. The postmark served as proof of a certain date, with obvious limitations: it did not uniquely identify the author, was not verifiable by third parties without opening and destroying the envelope, and depended on the physical preservation of a fragile object.

This protocol is the digital equivalent of that practice, with three structural differences: it identifies the author with state certainty through the CIE, is verifiable by anyone without opening anything, and does not degrade over time. It is not an invention: it is an adaptation to the digital world of a need Italian artists have known for decades.

The Aliexpress Model Applied to Art Authentication

Aliexpress did not create new logistics: it made existing logistics accessible. This protocol operates the same disintermediation in art authentication, using already existing tools — state digital signatures, ISO standard cryptographic algorithms, PDFs verifiable with free software.

The traditional chain is: artist → gallery → gallery certificate → collector. The SHA-256 + CIE Protocol reduces the chain to: artist → state-signed certificate → collector. The intermediary is not eliminated out of ideology: it is made optional because the tools to document without it already exist, are free, and in many contexts offer more verifiable documentary guarantees than an institutionally unsigned certificate.

Part 1 — Authenticating the Physical Work

This part covers what you do to the physical object before producing any document. The three elements that follow make each copy unrepeatable.

1.1 — The Handwritten Signature

Sign each copy by hand, with your full signature or a stable monogram. Standard position: back of the work, lower right. Use permanent ink (Pigma Micron, Staedtler Lumocolor). Do not use regular markers: they fade.

1.2 — Edition Numbering

Below the signature, write the copy number in the format:

→ N/TOTAL → e.g. 3/10 (third copy of an edition of ten)

→ A.P. → Artist's Proof (copy issued before the numbered edition begins)

Legal note

The edition declaration in the digitally signed certificate constitutes a private instrument with enhanced probative value under Legislative Decree 82/2005 (CAD) and EU Regulation 910/2014 (eIDAS), enforceable against third parties in accordance with the Civil Code. Anyone producing copies in excess of the declared number violates a legally documented obligation — not merely a moral promise.

1.3 — The Physical Seal

The seal is a unique visual element impressed by hand on each copy.

→ Custom engraved stamp: €15–30. The symbol must be yours.

→ Wax seal: each impression is unique due to micro-imperfections in the matrix.

→ Conceptual seal: derived from a text or theoretical system of yours.

Part 2 — The Certificate of Authenticity

Required Data

→ Author's full name (as on identity document)

→ Title of the work

→ Year of creation

→ Technique and materials (be precise: not "print" but "inkjet print on Hahnemühle 310g paper")

→ Exact format in cm

→ Copy number and total edition (e.g. 3/10 or A.P.) — binding declaration

→ Authentication method (signature + seal + hash + CIE)

→ SHA-256 hash of the source image file (see Part 8.1)

→ SHA-256 hash of the print photograph (see Part 8.0)

→ Edition clause: "The author undertakes not to produce further physical prints beyond those declared in this certificate. This undertaking, digitally signed with the CIE, constitutes a private instrument with enhanced probative value under Legislative Decree 82/2005 (CAD) and EU Regulation 910/2014 (eIDAS), enforceable against third parties in accordance with the Civil Code."

Recommended Data

→ Critical note (3 paragraphs: origin, what it does, where it fits in your work)

→ QR code linking to theoretical corpus, essay, or website

→ Theoretical reference with Zenodo DOI

→ Print materials: paper type, inks, colour profile

→ Conservation instructions: light, humidity, temperature

Output Format: PDF/A

Compose the document in Word, LibreOffice, or Canva. When exporting, use PDF/A-2b format. PDF/A is the ISO 19005 standard for long-term archiving: it guarantees the file remains readable and verifiable for 30–50 years. In LibreOffice: Export → PDF → General tab → tick "PDF/A-2b". In Word: Save As → PDF → Options → "ISO 19005-1 compliant (PDF/A)".

Part 3 — The Digital Authentication Chain

The five layers below apply exclusively to the certificate PDF. The order is binding: each layer must be executed after the previous one.

The 5 layers of the protocol

Layer 1 → Produce the PDF/A certificate
Layer 2 → Calculate the SHA-256 hash (+ SHA-512 optional)
Layer 3 → Document the hash in a second PDF/A
Layer 4 → Sign both PDFs with your CIE (read the FEA/FEQ note before proceeding)
Layer 5 → Send to yourself via PEC (mandatory step)

Layer 1 — The Certificate of Authenticity (PDF/A)

This is the document produced in Part 2. Do not proceed until the certificate is in its final form — correct text, approved layout, complete data. Any modification to the PDF after the hash has been calculated invalidates the entire chain.

Layer 2 — The SHA-256 Hash (and SHA-512)

SHA-256 is a cryptographic algorithm that converts any file into a unique 64-character string. If even one character of the PDF is modified, the string changes completely.

How to calculate the hash:

→ Windows: certutil -hashfile certificate.pdf SHA256
→ Mac: shasum -a 256 certificate.pdf
→ Linux: sha256sum certificate.pdf
→ Graphical tool: QuickHash GUI — quickhash-gui.org
→ Offline HTML tool: multi-algorithm generator in the protocol repository (sha_generator.html)

Layer 3 — The SHA Document

The calculated hash must be recorded in a separate document — the SHA Document — which becomes the second PDF/A in your documentation. It must contain:

→ Name of the certificate file and size in KB
→ Exact date and time of calculation
→ Complete SHA-256 hash string (64 characters)
→ Complete SHA-512 hash string (128 characters) — if calculated
→ Your full personal details

Technical note

The SHA Document itself is not hashed in the basic flow. Those wishing a cryptographically closed chain may calculate the hash of the SHA Document itself and annotate it in a third digitally signed text file — this is the advanced level described in Part 8.

Layer 4 — The CIE Signature

AES vs QES — What changes in court (read before signing)

Signing with the CIE via Firma4NG or CieSign produces an Advanced Electronic Signature (AES / FEA), not a Qualified Electronic Signature (QES / FEQ). This distinction has precise practical consequences in case of judicial dispute:

With AES (FEA): the burden of proof falls on the party presenting the document. If someone contests the signature, you must prove it was you who signed.

With QES (FEQ): the reverse burden of proof applies. The signature is presumed authentic until challenged by a formal complaint of forgery.

For the vast majority of works, the AES level is sufficient — the CIE + PEC combination is robust. For high-value works (> €5,000) or contexts of elevated contestation risk, consider a Qualified Electronic Signature (QES) via CNS token or professional smart card.

Required Tools

→ Your CIE with NFC chip
→ CIE PIN (delivered in a sealed envelope when the card was issued)
→ An NFC reader: USB smartcard reader (€10–15)
→ Firma4NG (PC — Windows/Mac/Linux): free AGID software, from agid.gov.it
→ CieSign (smartphone iOS/Android): official Ministry of the Interior app

Important distinction

Firma4NG and CieSign are two different tools for the same result. Firma4NG is the desktop solution, more stable for multiple signatures. CieSign is the mobile solution. Both produce PAdES signatures with identical legal value. If using a smartphone, use CieSign — not Firma4NG.

Procedure with Firma4NG (PC)

1. Install Firma4NG from agid.gov.it → Tools section. It is free and official.
2. Load both PDFs: the certificate of authenticity and the SHA Document.
3. Select PAdES — signature embedded in the PDF. Do not use CAdES (produces a separate .p7m file).
4. Hold the CIE near the NFC reader. Enter the PIN when prompted.
5. Verify the result: open the signed PDF in Adobe Acrobat Reader. A banner appears with your name and the date of signing.
6. Export the validation report (.p7s or verification PDF produced by Adobe Acrobat or Firma4NG) and keep it with the signed PDFs.

Layer 5 — The Timestamp via PEC (mandatory step)

PEC: a practical redundancy tool — not a qualified TSA

Certified email (PEC) creates proof of the document's existence at a date certified by a third-party operator accredited by AgID. It is an excellent tool for practical redundancy.

Technical clarification: PEC is not a TSA (Time Stamp Authority) accredited in the strict eIDAS sense. Its timestamp is that of the PEC operator, not of a qualified TSA. In judicial proceedings, a judge may request a qualified timestamp for "certain date" purposes under art. 2704 of the Civil Code.

For the majority of works, the CIE + PEC combination is amply sufficient. For high-value works, the advanced option is described below.

Immediately after signing with the CIE, send both signed PDFs to yourself via PEC.

→ Free or low-cost PEC providers: Aruba PEC, Libero Mail PEC, Legalmail — all AgID-accredited.

→ How: open your PEC client, attach both signed PDFs, send to your own PEC address. The delivery receipt is the independent timestamp.

→ Keep PEC receipts together with signed PDFs in the work's archive.

Advanced option: qualified timestamp (TSA)

For higher-value works or high-risk contestation contexts, a timestamp from an eIDAS-accredited TSA (e.g. Aruba TSA, InfoCert TSA) can be added. The service costs a few cents per document and produces a .tsr file that certifies the document's existence in fully qualified form under the eIDAS Regulation. A TSA does not replace PEC — it complements it.

Part 4 — Extension to Art Dealers

4.1 — The Dealer as Node Validator

The dealer does not create the certificate from scratch, but digitally signs — with their own CIE or professional digital signature — the certificate already issued by the artist. The resulting PDF contains two distinct signatures:

→ The author's signature: certifies the origin and authenticity of the work.

→ The dealer's signature: certifies the chain of custody and transfer of ownership.

This double signature is cryptographically verifiable by anyone, at any time, without contacting either the artist or the dealer.

4.2 — Reducing Due Diligence Costs

→ Every work has an independently verifiable documentary chain.

→ In case of fire, theft, or loss of physical documents, the signed digital archive has full legal value for insurance purposes.

→ Authenticity disputes are resolved mathematically: either the hash matches, or it does not.

4.3 — Managing Editions and Print Runs

The dealer can issue a digitally signed Edition Report listing the hashes of all individual certificates in a series. This document certifies that the list is complete and immutable as of the date of signing.

4.4 — Transparency as a Sales Argument

> "Dear client, this photograph is accompanied by a security protocol that requires no intermediaries. Here is the SHA-256 hash: you can verify at any time that this is the only certificate issued by the author, on that date, with that state signature. Few solutions available today offer a comparably verifiable documentary chain."

Part 5 — Summary of the Complete System

Physical Work

→ Handwritten signature + Numbering (N/TOTAL or A.P.) + Physical seal
→ Signed photograph of the print (see Part 8.0)

Digital Documentation — 5 Layers

→ Layer 1: PDF/A certificate of authenticity (data + critical note + QR + edition clause)
→ Layer 2: SHA-256 hash of the PDF → immutable string (+ SHA-512 optional)
→ Layer 3: SHA Document → second PDF/A with documented hash
→ Layer 4: CIE signature on both PDFs → identity certified by the State
→ Layer 5: PEC sending of signed PDFs → certified independent timestamp

What the Collector Receives

→ The physical work with signature, number, and seal
→ The PDF/A certificate of authenticity (CIE-signed)
→ The SHA Document PDF/A (CIE-signed)
→ The transfer certificate (signed by both parties — see Part 13)
→ The assurance that none of the documents can be altered without detection

Operational Checklist — Before Each Delivery

Physical work:
- ☐ Handwritten signature applied
- ☐ Copy number written by hand (N/TOTAL or A.P.)
- ☐ Physical seal impressed
- ☐ Print photograph taken and hashed (Part 8.0)

Digital documents:
- ☐ Certificate of authenticity in PDF/A — complete data + edition clause
- ☐ SHA-256 hash of source image file included in certificate
- ☐ SHA-256 hash of print photograph included in certificate
- ☐ Critical note included
- ☐ QR code to theoretical corpus
- ☐ SHA-256 hash calculated on final PDF (+ SHA-512 optional)
- ☐ SHA Document produced
- ☐ Both PDFs signed with CIE
- ☐ Validation report .p7s exported and kept
- ☐ Both PDFs sent to yourself via PEC — receipts kept
- ☐ Transfer certificate generated and signed by both parties

Personal archive:
- ☐ Copy of all signed PDFs saved locally + cloud (3 copies, 2 media, 1 offsite)

☐ Work register: title, copy, recipient, date, hash

Part 6 — Letter to the Collector

> Dear Collector, With the work you have received, you will find a set of digital documents. I know they may seem like a lot, and that some terms — hash, digital signature, SHA-256 — sound technical and unfamiliar. I want to explain in plain language what they are, why I care that you keep them, and what they guarantee. You have received five things: → The physical work. Signed by hand, numbered, sealed and with the photograph of the print. It is unique in the physical world. → The certificate of authenticity. A PDF document explaining who made this work, when, with what technique, how many copies exist. → The SHA-256 document. The 64-character string — the digital fingerprint of the certificate. → The digital CIE signature. Both PDFs have been signed with my Electronic Identity Card, with full legal value in Italy and throughout the EU. → The transfer certificate. It documents that this specific copy passed to you, on this date, from me. If you sell it on, you can add a second transfer — the chain remains verifiable forever. If in twenty years someone asks you to prove this work is authentic, all you need to do is open those PDFs in Adobe Reader. No need to contact me, no gallery, no intermediary needed. I ask only one thing: keep these files on at least three separate media — a hard drive, a USB stick, a cloud service. Sincerely, Dmitrij Anatolievic Musella

Part 7 — Objections and Responses

Objection 1 — The Traditional Critic

> "All this procedure kills the poetry of exchange."

Technology does not replace art: it protects it over time. A restorer consolidating a canvas with chemical resins is not killing the painting — they are ensuring it exists in a hundred years. The digital chain is invisible at the moment of giving, but becomes fundamental twenty years later.

Objection 2 — The Gallery Owner

> "An artist who issues their own certificates looks like an amateur."

A certificate produced with the CIE has a verifiable documentary chain that many traditional gallery certificates, not digitally signed, cannot offer. The difference is that the CIE is issued by the Ministry of the Interior. A gallery issues certificates on headed notepaper.

Objection 3 — The IT Specialist

> "Who guarantees the file was not modified before the calculation? And in twenty years, who will be able to read PAdES?"

SHA-256 is a universal ISO standard. PAdES is the European standard for digital signatures in PDFs, independent of any software. PEC adds a timestamp from an accredited third-party operator. PDF/A guarantees long-term readability under ISO standards.

Objection 4 — The Non-Technical Collector

> "Why do I need to keep 5 files and install programmes?"

The nice sheet with stamp and signature exists: it is the printed paper certificate. The digital files are like the black box of an aircraft: we hope never to need to open it, but we are glad it is there. Verification requires opening a file in the free Adobe Reader.

Objection 5 — The Jurist

> "They created the certificate, calculated the hash, and signed everything themselves. Judge and party."

The CIE signature is not self-certification: the private key is held by the Italian State, not the artist. The PEC timestamp adds a certified third party.

AES vs QES — Clarification

The CIE signature via Firma4NG/CieSign is an Advanced Electronic Signature (AES/FEA). Those requiring the highest probative level may use a Qualified Electronic Signature (QES) via CNS token or accredited smart card — which produces the reversal of the burden of proof (presumption of authenticity until a formal complaint of forgery).

See Layer 4 for a full explanation of the procedural implications.

Objection 6 — The Subtlest One

> "If everyone adopts this method mechanically, it becomes an empty ritual."

This is the most honest critique, and it must be accepted. The system works well only if those using it have something real to document. A SHA-256 hash cannot create value where none exists: it can only protect value that already exists.

Part 8 — The Closed Circle: Synchronising Physical Work and Digital Certificate

8.0 — The Hash of the Print Photograph

Problem addressed in this paper

This section addresses the most important structural problem: earlier elaborations certified the digital file and the document, but did not cryptographically anchor the physical print to the certificate. The hashed photograph bridges this gap.

In the same session in which you sign the certificate with the CIE, follow this procedure for each physical copy:

7. Take a high-resolution photograph of the physical print (front and back), including in the frame a sheet with the date handwritten, the copy number, and your handwritten signature.
8. Calculate the SHA-256 hash of this photographic file.
9. Insert this hash string in the certificate of authenticity, in the field "SHA-256 hash photograph print [N/TOTAL]".
10. Keep the photographic file in the work's archive together with the signed PDFs.

What this step guarantees:

→ Cryptographically links each physical copy to its digital certificate. Two different prints, even visually identical, produce photographs with different hashes.

→ Provides a documented and dated visual proof of the copy, signed in the same CIE session as the certificate.

→ In case of dispute ("this print is not the original"), there exists a signed, hashed, and dated photograph of the copy at the moment of its creation.

Clarification: verification for the collector remains based on the PDF hash

The hash of the photograph serves as an archival anchor, not as a direct verification tool for the collector.

If the collector were to rephotograph the work and compare the new hash with the one in the certificate, they would not find a match: this is normal and expected, since digital photographs produce different hashes even of the same subject. Verification remains based on the hash of the certified PDF, which is deterministic and invariable.

The archived photograph serves the artist as documentary evidence in case of judicial dispute: it demonstrates that that specific copy existed in that form, on that date, signed by that person.

8.1 — The Hash of the Source Image File

Before printing the work, calculate the SHA-256 hash of the original high-resolution image file (the TIFF or JPEG master). Insert this string in the certificate of authenticity. This creates the mathematical link between the printed file and the document that describes it.

→ Source file conservation: the source TIFF is the origin point of the chain. Keep it on at least three separate physical media. If the source file is lost, verification of the source hash becomes impossible.

8.2 — The Edition Declaration as a Legal Commitment

The certificate must contain the explicit clause indicated in Part 2. By signing with the CIE, the commitment becomes a private instrument with enhanced probative value. It is no longer merely a moral promise: it becomes a documented and verifiable obligation. Anyone producing copies in excess violates this obligation in a documented and mathematically provable manner.

8.3 — Technical Metadata and Link to the Theoretical Corpus

→ Production tool: specify the tool used (camera, lens, software, manual technique), execution conditions, relevant technical parameters. Someone using a dichroic prism will specify model and configuration; a painter will specify support and medium; a sculptor will specify material and process. The goal is to make the work reproducible in description even by someone who was not there.

→ Corpus reference: if the work belongs to a cycle, series, or theoretical system, cite it here with a verifiable reference. It may be an essay published in a journal, a text on Medium or Substack, a catalogue entry, or a document deposited on Zenodo with a DOI. A DOI (Digital Object Identifier) is a unique permanent code assigned to a digital document by accredited archives such as Zenodo, managed by CERN with European funding. A DOI guarantees that the link to the document never changes over time, unlike an ordinary URL. If you have no published theoretical corpus, omit this field: it is not mandatory. A work does not need an essay to be authentic.

→ Materials and conservation: for works on paper, specify paper type and weight, ink or medium type, colour profile. For digital works, specify the master file format and print conditions. Add conservation instructions: light, humidity, temperature. These data are irrecoverable after the loss of the physical work.

8.4 — The QR Code as a Physical-Digital Bridge

Insert a QR code in the paper certificate pointing to the digitally signed PDF, ideally hosted on Zenodo.

> **Note on Zenodo dependency**
>
> Zenodo is managed by CERN with EU funding and has excellent stability, but remains a third-party platform. For redundancy, also host the signed PDFs on a personal repository. The QR code can point to Zenodo as primary, with an alternative URL indicated in the text.

8.5 — The Complete Closed Circle

→ Source image file → SHA-256 hash inserted in the certificate PDF

→ Physical print → signed photograph → SHA-256 hash inserted in the certificate PDF

→ Certificate PDF → SHA-256 hash inserted in the SHA Document

→ Both PDFs → CIE-signed (state identity)

→ Both PDFs → sent via PEC (independent timestamp)

→ Validation report .p7s → kept with signed PDFs

→ Paper certificate → QR code pointing to signed PDF on permanent repository

→ Physical work → handwritten signature, numbering, physical seal

Part 9 — Towards a Standard: Proposal for Institutional Adoption

9.1 — Institutional Recipients

→ AGID (Italian Digital Agency) — manages CIE and Firma4NG
→ ANGAMC (National Association of Modern and Contemporary Art Galleries)
→ FIMA (Italian Federation of Art Dealers)
→ Fine Arts Academies: adoption as a teaching protocol
→ European Commission within the eIDAS 2.0 revision

9.2 — The Artist's Role in Setting Standards

Historically, art market standards have been defined by institutions. This paper reverses the direction: it is an independent artist proposing a technical standard to the market, founded on existing tools and validated by practical experience. The Italian art market could benefit from more structured and verifiable documentary methods. This paper proposes one possible approach, open to critical contribution from anyone wishing to refine it.

Part 10 — Blockchain/NFT vs SHA-256 + CIE Protocol

The difference is not technical, it is one of identity. A blockchain certifies that something exists at a given moment. It does not know who you are. The CIE is issued by the Ministry of the Interior, knows exactly who you are, and certifies it with a private key held by the State.

Criterion	Blockchain / NFT	SHA-256 + CIE
Signer identity	Anonymous (wallet)	State-certified (CIE)
EU legal value	None direct	Full (eIDAS, AES)
Cost	Gas fees + platform	Free
Verification	Requires technical expertise	Free Adobe Reader
Third-party dependency	Yes (platform may close)	No
Obsolescence risk	High	Low (SHA-256 is ISO)
Registered identity	No	Yes
Ownership transfer	Token — not a person	Certificate signed by both parties
Provenance chain	Token only	CIE-signed verifiable documents
Accessible to traditional collector	No	Yes

Part 11 — Long-Term Preservation of Digital Documents

11.1 — The Problem of Format Obsolescence

The correct question is not "does this file exist today?" but "will this file be readable and verifiable in 30 years?" The protocol addresses this at three levels:

→ PDF/A: the ISO 19005 standard for long-term archiving. Self-contained, includes all fonts and metadata needed.

→ SHA-256 and SHA-512: algorithms of the ISO/IEC 10118-3 standard. Not proprietary formats.

→ PAdES: the European ETSI EN 319 102 standard for digital signatures in PDFs.

11.2 — The 3-2-1 Rule

→ 3 copies of the file (original + 2 backups)

→ 2 different physical media (e.g. external hard drive + USB stick)

→ 1 copy offsite (cloud service, or physical copy in a different location)

11.3 — The Permanent Repository

Zenodo (zenodo.org) is the recommended permanent repository. Managed by CERN with EU funding, it offers permanent DOIs and commits to long-term preservation. Depositing on Zenodo does not replace local conservation — it complements it.

11.4 — The CIE Expires: What Happens to Old Signatures?

Digital signatures applied with a valid CIE remain verifiable even after the card expires. Verification is based on the certificate that was valid at the time of signing, not the current one. Adobe Reader shows the signature with the original date and the certificate status at the time of signing ("signature valid at the time of application").

Additional recommendation: at the time of signing, export the validation report (.p7s or verification PDF produced by Adobe Acrobat or Firma4NG) and keep it with the signed PDFs. This file contains proof of the certificate's validity at the time of signing, independently of any future software.

11.5 — Conservation Instructions to the Collector

→ Keep files on at least three separate media (3-2-1 rule)

→ Do not rename the files: the file name is part of the documentary chain

→ In case of sale, transfer the digital files together with the physical work and generate a transfer certificate (see Part 13)

Part 12 — The Hash Generator: Open Source Tool

A multi-algorithm generator is available as an HTML web page that works completely offline. It is a single HTML file downloadable and openable in any modern browser without an internet connection and without installing anything. It produces:

→ SHA-256 hash (64 characters) — mandatory standard of the protocol
→ SHA-512 hash (128 characters) — professional option
→ BLAKE2b — optional additional layer (see Part 12.1)
→ Pre-formatted text ready to paste into the SHA Document (Layer 3)

12.1 — The Algorithms: SHA-256, SHA-512, and BLAKE2b

SHA-256 is the mandatory standard of this protocol: 64 hexadecimal characters, adopted by ISO, NIST, recognised by European jurisprudence and eIDAS regulations. It is the legal reference standard: any certificate must include the SHA-256 hash as its primary element.

SHA-512 is the professional variant: 128 characters, greater computational robustness. It belongs to the same SHA-2 family and is equally recognised. It is recommended as a secondary hash for high-value works.

BLAKE2b is an optional algorithm for those seeking algorithmic diversification. It does not belong to the SHA-2 family: designed in 2012 by an independent team (Aumasson, Bernstein, Lange, and others), it is faster than SHA-256 on 64-bit hardware and has no known vulnerabilities. It is used in high-security contexts such as WireGuard and Argon2.

> Hierarchy to respect: SHA-256 is always mandatory. SHA-512 is recommended as a secondary hash. BLAKE2b is an optional additional layer. A document containing only BLAKE2b, without SHA-256, may encounter resistance in judicial or institutional contexts where SHA-256 is the expected standard. Always use BLAKE2b in addition, never as a substitute.
>
> MD5 and SHA-1 are deliberately excluded: cryptographically compromised, invalid for any security application.

12.2 — No Data Sent to Any Server

Calculation takes place entirely in the browser, on the user's local device. No file is ever sent to external servers. The source code is public and verifiable.

12.3 — The Collector Verifier (Green/Red Light)

To reduce technical friction for non-expert collectors, the protocol includes a second offline HTML page: the interactive verifier (sha_verifier.html). It is a single HTML file that the artist creates once and delivers to the collector alongside the signed PDFs. The collector opens it in their browser (Chrome, Firefox, Safari, Edge), without an internet connection and without installing anything.

It works in three steps:

11. The collector drops the signed PDF into the HTML page.
12. The page automatically calculates the SHA-256 hash of the file.
13. It compares the calculated hash with one manually entered by the collector (copied from the certificate). It shows a green light if they match, red if they do not.

Zero backend. Zero installations. Zero technical expertise required.

Project philosophy

The verifier is deliberately separate from the generator. They are two distinct tools for two distinct audiences: the generator is for the artist (creation phase), the verifier is for the collector (verification phase). Both work offline, without servers, without data sent to third parties. Both are single downloadable HTML files. Both are open source and verifiable.

Part 13 — Transfer of Ownership

Why this part exists

This part addresses a structural problem: previous elaborations covered the certification of the work. They did not cover what happens when the work changes hands. Part 13 solves this without adding platforms, services, or costs.

13.1 — The Problem of the Second Transfer

The original certificate is in the artist's name. When the collector sells the work ten years later, how does the new owner demonstrate it is authentic? The certificate says nothing about them. The provenance chain is broken.

This is the gap that the traditional market solves through the gallery: the gallery certificate follows the work because the gallery guarantees its provenance with its own institutional reputation. The protocol solves the same problem without the gallery, through documents digitally signed by both parties, mathematically linked to the artist's original chain. No platform, no intermediary, no additional cost.

13.2 — The Transfer Certificate

When the work changes hands, a second document is generated: the Transfer Certificate. It is a simple PDF that anyone can complete in Word or LibreOffice, export as PDF/A, and sign. It requires no specialised software. It requires no notary. No gallery. It contains:

→ SHA-256 hash of the original certificate of authenticity (cryptographic link to the chain)

→ Work data: title, copy number, year

→ Transferor's details: name, surname, tax code (or passport number for non-Italian parties), date

→ Recipient's details: name, surname, tax code or passport number

→ Date of transfer

→ Price (optional)

→ Transferor's declaration: "I certify that I am transferring ownership of this copy on the date indicated, and that no physical copies exist beyond those declared in the original certificate"

13.3 — The Signing Procedure

14. The transferor signs the Transfer Certificate with their CIE (or scanned handwritten signature if they do not have one).
15. The recipient signs in turn with their CIE (or scanned handwritten signature).
16. Both send the signed document via PEC to the other party and to themselves.
17. The Transfer Certificate is kept together with the original Certificate of Authenticity and the SHA Document.

Nota pratica

Legal note — Simple Electronic Signature (SES): a scanned handwritten signature constitutes a Simple Electronic Signature under the eIDAS Regulation. Unlike a CIE signature (AES), its probative value is freely assessable by a judge in case of dispute, under art. 2702 of the Civil Code. The cryptographic chain of hashes remains intact, but the recipient's identity loses the state guarantee of

the CIE. Using the CIE by both parties is strongly recommended to maintain the AES level and full probative effectiveness of the document.

13.3b — Foreign Collectors and Non-Italian Buyers

The protocol in its complete form was designed for the Italian legal context. When one party is foreign, the protocol adapts differently:

→ EU buyers. The eIDAS Regulation (910/2014) guarantees mutual recognition of qualified electronic signatures among all member states. A German, French, or Spanish buyer can sign the Transfer Certificate with their national digital signature with full legal validity in Italy. The Italian tax code can be replaced by the national identity document number.

→ Non-EU buyers (USA, Japan, others). The artist signs the Transfer Certificate with their CIE and sends via PEC: this part remains unchanged and has full legal validity in Italy. The buyer signs with a handwritten signature, or with their country's digital signature if recognised. The passport number replaces the tax code. For significant value transactions, an apostille on the buyer's signature can be used, extending international recognition under the 1961 Hague Convention.

→ Declared limit. The protocol in its optimal form is designed for the Italian legal context and, by extension, the European one. Transactions with buyers from countries with very different legal systems (common law, Islamic law, etc.) may require specific adaptations outside the scope of this paper. In such cases, consulting a legal professional with experience in private international law and art market transactions is advisable.

13.4 — Subsequent Transfers

Each transfer generates a new Transfer Certificate. Each document contains the hash of the previous one, creating a mathematically verifiable provenance chain, independent of any platform:

→ Original Certificate (artist → collector A) — CIE-signed by the artist

→ Transfer Certificate 1 (A → B) — hash of Original, signed by A and B

→ Transfer Certificate 2 (B → C) — hash of TC1, signed by B and C

Anyone who has the work can at any time verify the entire chain by opening the PDFs in Adobe Reader and checking signatures and hashes. No need to contact anyone. No platform needed. No gallery needed. The memory of the work is in the files, and those files belong to the artist, the collector, and anyone who owns the work in future.

Each transfer of ownership is an act of documented responsibility: the transferor signs that they are transferring, the recipient signs that they are receiving. There is no possible ambiguity at the documentary level. There is no third party to delegate the memory of the work to.

13.5 — Transfer Certificate Template

CERTIFICATE OF TRANSFER OF OWNERSHIP
SHA-256 + CIE Protocol

Work: [Title] · Copy: [N/TOTAL] · Year: [Year]
Hash of original certificate (SHA-256):
[64 hex characters]

TRANSFEROR
Name: _____ Tax code / Passport: _____
Signature: _____ Date: _____

RECIPIENT
Name: _____ Tax code / Passport: _____
Signature: _____ Date: _____

Price (optional): _____

The transferor declares they are transferring exclusive ownership of this copy on the date indicated, and that no physical copies exist beyond those declared in the original certificate.

Sign digitally with CIE (PAdES) and send via PEC from both parties.

Pilot Case — Il Ticket Infinito (March 2026, Pre-Protocol)

Methodological note: this is the first real application of the protocol, carried out in March 2026 during an experimental phase preceding the version described here. Some elements were not yet formalised at the time of the application, in particular Layer 5 (PEC) and the source file hash. The gaps in this pilot application contributed to defining the requirements of the current protocol. What follows is not a demonstration of a perfect system, but proof that the protocol works in the real world, with a professional art market operator, in the ordinary conditions of an independent artist.

The Context

On 17 March 2026, the author donated the Artist's Proof (A.P., copy 1/1 preceding the edition) of the work "Il Ticket Infinito" to Pietro Mingione, founder and director of Station Gallery. The donation took place in the context of a studio visit and a meeting with the gallery's curator, who had previously evaluated the author's photographic work positively.

"Il Ticket Infinito" is a digital illustration on Fuji photographic paper, A4 horizontal format, with frame included. The work belongs to Chapter 4 of the III Esalogia of the essay "Produ-nied" and is inseparable from the theoretical corpus that generated it: the certificate of authenticity contains two QR codes linking directly to the complete essay on Medium and to the GLYPH9 cipher-writing algorithm, derived from the essay "La Freccia Invertita".

The Documentary Chain Produced

→ Layer 1: PDF/A certificate of authenticity with complete data, critical note for the collector, QR codes to theoretical essays, GLYPH9 seal, handwritten signature. Timestamp: 17/03/2026 22:37:11.

→ Layers 2 and 3: SHA-256 hash calculated at 22:23:54. Value: a285d9fea10f6098cedc78197965b3c936ff74da19de6e07b598ea2867585873. SHA Document CIE-signed at 22:38:37.

→ Layer 4: CIE digital signature (PAdES) on both PDFs, with complete author details (name, surname, date and place of birth, tax code MSLDTR88C09Z154K) verifiable in Adobe Acrobat Reader.

→ Layer 5 (PEC): not executed in this first application. The gap is known and documented. In subsequent applications, Layer 5 is mandatory.

Proof of Delivery

The delivery is documented by a screenshot of the WhatsApp conversation with Pietro Mingione Station Gallery, timestamped between 22:31 and 22:36 on 17/03/2026, coinciding exactly with the moment of production of the signed PDFs. The screenshot shows the explanation of the protocol to the recipient in real time and their immediate reaction. The recipient is a professional art market operator: the donation occurred in the context of an artist-gallery relationship, with a curator involved in the preliminary evaluation of the work.

Documented Gaps and the Correction Path

Three elements of the current protocol are absent from this first application: PEC (Layer 5), the source image hash, and the recipient's name in the signed PDFs. These gaps are documented here deliberately. A paper that conceals the gaps of its first real application is not a scientific paper. The documentation of gaps is simultaneously proof that the improvement

process is underway: the protocol has already incorporated corrections in this paper based on analysis of this concrete case.

What It Demonstrates, in Its Imperfection

The case demonstrates four things that no theoretical version of the protocol could demonstrate alone. First: the protocol is executable in real conditions, the same evening as a donation, without specific preparation. Second: a professional art market operator received and understood the system without prior technical training. Third: the CIE signature works as intended, with complete and verifiable personal details. Fourth: the link between physical work, theoretical corpus, and digital documentation is achievable coherently through the GLYPH9 seal, QR codes, and critical note included in the certificate.

This is not a finished system. It is a system that works, that improves with each iteration, and that documents its own improvement. The distance between the first application and the protocol described in this paper is the most concrete proof that the method is alive.

Resources

→ Firma4NG (CIE digital signature on PC): agid.gov.it → Tools
→ CieSign (CIE digital signature on smartphone): App Store / Google Play
→ Multi-algorithm hash generator (offline, open source): sha_generator.html (protocol repository)
→ Collector verifier with green/red light: sha_verifier.html (protocol repository)
→ QuickHash GUI (graphical hash calculation): quickhash-gui.org
→ Digital signature verification: Adobe Acrobat Reader (free)
→ Zenodo (permanent repository with DOI): zenodo.org
→ Aruba TSA (qualified eIDAS timestamp): pec.aruba.it
→ InfoCert TSA (qualified eIDAS timestamp): infocert.it
→ eIDAS Regulation: EUR-Lex 910/2014
→ CIE verification: cieid.interno.gov.it
→ PDF/A standard: ISO 19005 — available from iso.org

Appendix A — Critical Analysis: Twelve Identified Issues

Issue 1 — Firma4NG ≠ smartphone

Problem	Problem: An earlier elaboration implied that one could sign from mobile with Firma4NG. Incorrect: Firma4NG is for PC only.
Solution	Solution (earlier versions): Explicit distinction between Firma4NG (PC) and CieSign (smartphone). No residual ambiguity.

Issue 2 — CIE signature timestamp is not always a TSA

Problem	Problem: An earlier elaboration stated the timestamp was that of the Ministry, not retroactively modifiable. Technically imprecise.
Solution	Solution (earlier versions): PEC becomes the mandatory 5th layer. Language on the timestamp corrected.

Issue 3 — AES vs QES: legal robustness

Problem	Problem: An earlier elaboration equated the CIE signature to an authenticated private deed in absolute terms, without distinguishing AES from QES.
Solution	Solution (earlier versions): AES/QES distinction added. Solution in this paper: The distinction is now at the heart of Layer 4, before proceeding to sign, with explicit explanation of the burden of proof.

Issue 4 — SHA Document not in the hash chain

Problem	Problem: The SHA Document was not itself hashed. The cryptographic chain was incomplete.
Solution	Solution (earlier versions): Technical note added. Advanced level (hash of SHA Document) described as an option in Part 8.

Issue 5 — QR Zenodo: undeclared third-party dependency

Problem	Problem: Earlier versions criticised blockchain for third-party platform dependency, then proposed Zenodo without acknowledging the contradiction.
Solution	Solution (earlier versions): The contradiction is declared. Proposal for redundancy: Zenodo as primary + personal URL.

Issue 6 — CIE expires: what happens to old signatures

Problem	Problem: Earlier versions did not address the expiry of the CIE (10 years) and the validity of signatures already applied.
Solution	Solution (earlier versions): Section 11.4 added. Solution in this paper: recommendation to export the .p7s validation report at the time of signing.

Issue 7 — Absence of physical-digital link

Problem	Problem: The protocol certified the digital file and document, but did not cryptographically anchor each specific physical copy.
Solution	Solution in this paper: Part 8.0 — the hashed photograph of the print, taken in the same CIE signing session.

Issue 8 — Absence of ownership transfer

Problem	Problem: The protocol covered creation of the work, not its circulation. At the second sale, the provenance chain was broken.
Solution	Solution in this paper: Part 13 — the Transfer Certificate creates a mathematically verifiable provenance chain, signed by transferor and recipient with CIE, without third-party platforms.

Issue 9 — PEC as TSA equivalent

Problem	Problem: PEC was presented as a certified timestamp without distinguishing it from an eIDAS-accredited TSA.
Solution	Solution in this paper: Layer 5 — clarified that PEC is an excellent practical redundancy tool, not a qualified TSA in the strict sense. Optional path to accredited TSA (Aruba, InfoCert) added for high-value works.

Issue 10 — Imprecise legal terminology

Problem	Problem: The expression "authenticated private deed" presupposes notarial authentication (art. 2703 c.c.). The CIE (AES) does not constitute this act.
Solution	Solution in this paper: Replaced throughout with "private instrument with enhanced probative value under Legislative Decree 82/2005 (CAD) and EU Regulation 910/2014 (eIDAS)".

Issue 11 — Ambiguity on photograph verification

Problem	Problem: The description in 8.0 implied the collector could verify the photograph by comparing hashes. Misleading: rephotographing always produces different hashes.
Solution	Solution in this paper: Explanatory box added in 8.0. The photograph is an archival anchor for the artist. The collector's verification remains based on the PDF hash.

Issue 12 — Technical friction for the collector

Problem	Problem: Verification required opening Adobe Reader and interpreting a signature banner. A practical obstacle for non-technical users.
Solution	Solution in this paper: Part 12.3 — offline HTML verifier with green/red light. Drop the PDF, paste the hash, get the answer. Zero technical expertise required.

Appendix B — Legal Blind Spots: Analysis and Limits of the System

Blind Spot 1 — Self-Certification and Absence of a Third-Party Register

The system does not include a public register of works. The hash proves the certificate was not altered after signing, but does not prove the certificate was produced before delivery of the work, nor that the work physically exists.

Mitigation: PEC creates proof of the document's existence at a date certified by a third party. For a further level: depositing on Zenodo before delivery creates a public and verifiable timestamp.

Residual limit: this system is honest by design, not by structure. It is the limit of any self-certification system — including traditional gallery certificates.

Blind Spot 2 — Physical Uniqueness of Prints

The hash certifies the digital certificate and the source file. It does not physically certify that the work has been printed in the number of copies declared.

> Declared limit: the guarantee of physical uniqueness depends on the artist's honesty. The protocol cannot replace human trust — it can only document it and make its violation legally and mathematically demonstrable. It is the same logic as a notarial deed: it does not prevent a seller from selling the same property twice, but whoever does so commits a provable offence.

Blind Spot 3 — Long-Term Preservation

Addressed in Part 11. Solution: PDF/A format, 3-2-1 rule, conservation on at least three separate physical media, .p7s report, explicit instructions to the collector.

Note on direct responsibility: this protocol does not delegate preservation to any platform. Zenodo may be used as an optional additional layer, but it is not an art work archive, and depending on it as a primary solution would contradict the philosophy at the basis of this study. The principle is the opposite: the artist takes direct responsibility for preserving their work and its documentation. They do not delegate it to a gallery, a blockchain, or an external server. They keep it, manage it, transmit it.

Blind Spot 4 — Transfer Chain Depends on Cooperation

The Transfer Certificate works well if parties cooperate. If the transferor sells the work without generating a transfer certificate, the chain breaks.

Mitigation: the clause in the original certificate can include the commitment to generate a transfer certificate at each change of ownership. This commitment, signed with the CIE, is legally binding.

Residual limit: no documentary system fully resolves the problem of bad-faith parties. This applies to notarial deeds too.

Final Note

This system does not replace galleries when galleries are the right context. It replaces the vacuum: those cases where the artist sells, donates, or exchanges a work without anyone bothering to document it seriously.

This paper extends the protocol to cover the entire lifecycle of the work: from creation to first sale, and from there to any subsequent transfer. The provenance chain can be documented mathematically, without intermediaries, using free tools legally valid throughout the European Union.

> "I do not do this to prevent forgers. I do it because I believe every work deserves an indestructible memory — a bridge between the physical gesture and the time to come."

Invitation to Critical Review

The author is a photographer and independent researcher, not a jurist or a cryptographer. The legal interpretations in this paper — in particular those concerning the probative value of AES, the value of PEC as a timestamp, and the applicability of the eIDAS Regulation to art ownership transfers — have not been validated by a legal professional specialising in digital law or art market law.

Similarly, the cryptographic observations on the longevity of the SHA-256 and BLAKE2b algorithms represent a synthesis of available literature, not an original technical opinion. The protocol as a whole has not undergone formalised academic peer review.

This paper is a working proposal, not a definitive standard.

Open to Collaboration

If you have read this paper and have something to say — a correction, an objection, an integration, a parallel case from your own practice — write to me. I am not looking for validation: I am looking for interlocutors willing to reason about the problem. The protocol improves to the extent that it finds people willing to test it seriously.

→ You are a jurist and want to flag an error in the legal interpretations? Write to me.

→ You are a cryptographer and have objections to the algorithm hierarchy? Write to me.

→ You are an artist or art dealer and have applied or want to apply the protocol? Tell me about the experience.

→ You are an academic or archivist and want to propose a comparison with existing standards? Write to me.

→ You have a real contestation case in which this system could have been used? Let us document it together.

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